ITC327W GROUP ASSESSMENT 1
MEMBER 7 – TECHNICAL DEVELOPMENT SUBMISSION

Project: Click Computer and Accessories
Former project name: ServiceHub IT
Prepared for submission to the Group Leader
Date: 26 August 2026

1. Purpose of this Submission

This document consolidates Member 7's technical-development responsibilities and the evidence package to be incorporated into the group's ITC327W Assessment 1 submission. The work is based on the ServiceHub IT technical foundation, adapted to the stakeholder's requested name, simpler client experience, device-repair workflow and 24/7 client-management concept.

2. Member 7 Role

Role: Technical Developer / Software Developer

- Develop and maintain the working web application.
- Implement authentication and role-based access for Client, Technician and Admin users.
- Develop client repair-request workflows and device/issue selection.
- Develop technician assignment and repair-status workflows.
- Integrate the application with Supabase for authentication/data/storage.
- Develop the Flutter client/mobile version of the platform.
- Perform technical testing, debugging and corrective changes.
- Maintain source code and provide GitHub evidence of individual contribution.

3. Technology Stack

Technology	Use in Project	Evidence
ASP.NET Core 8 MVC	Web application structure, controllers, views and server-side workflows	Source code / running application
C#	Application logic, controllers, models and business rules	Source code / Git commits
Supabase	Authentication, database, storage and backend services	Database/configuration evidence
JavaScript	Client-side interactions and notification/UI behaviour	Source code / screenshots
HTML/CSS	Web interface and responsive presentation	Views / screenshots
Flutter / Dart	Mobile client application	Flutter source / screenshots
Git / GitHub	Version control and contribution history	Commit screenshots / repository
PDF/Document handling	Existing ServiceHub IT technical capability where applicable	Relevant source code

4. Technical Scope to be Implemented

4.1 Client Functions

- Login / authentication
- Simple client dashboard

- Select device: Cellphone, Laptop, Printer or TV
- Select repair category: Hardware or Software
- Describe the problem and submit a repair request
- Track repair/request status
- Receive system notifications
- Logout

4.2 Admin / Support Functions

- View incoming client requests
- Manage clients
- View repair requests
- Assign a technician
- Monitor repair status
- Manage technician workload
- Use the system as the central management interface
- Support the stakeholder's requirement for active human client management

4.3 Technician Functions

- View assigned repair requests
- Review client/device/problem details
- Update repair status
- Record progress
- Mark a repair complete

5. ServiceHub IT Technical Foundation

The technical implementation is based on the ServiceHub IT . capabilities
include role-based dashboards, authentication/routing, ticket creation, administrator assignment of technicians, technician dashboards, client/employee ticket views, due dates and notifications.
the project is presented as Click Computer and Accessories.

- AuthController.cs – authentication and login routing
- AdminController.cs – administration and technician assignment
- EmployeeController.cs – adapted to Client functionality
- TechnicianController.cs – technician workflow
- Index.cshtml and related Views – interface pages
- Supabase database integration and Row Level Security considerations
- Notification centre – unread notification handling and workflow notifications

6. Proposed Technical Architecture

Client interfaces (web and Flutter) communicate with the application/backend services and Supabase. Supabase provides authentication and persistent data services. The web application uses ASP.NET Core 8 MVC with C#, while HTML/CSS/JavaScript support the presentation and client-side behaviour. GitHub provides version control.

Layer	Technology	Responsibility
Presentation	HTML, CSS, JavaScript, Flutter/Dart	Client, Admin and Technician interfaces
Application	ASP.NET Core 8 MVC + C#	Controllers, workflows, validation and role logic
Backend/Data	Supabase	Authentication, database and storage
Version Control	Git/GitHub	Source control and individual contribution evidence

7. Core Workflow

1. Client authenticates.

2. Client selects a device and identifies hardware/software repair type.

3. Client submits a repair request.

4. Request is stored in the backend/database with status Open.

5. Admin/support staff reviews the request.

6. Admin assigns an available technician; status changes to Assigned.

7. Technician views the assigned repair and updates progress/status.

8. Client receives/reads notifications and tracks the repair.

9. Technician completes the repair and the request is marked Complete/Resolved.

8. Database / Data Model

The following logical entities should be represented in the database and aligned with the actual Supabase schema used by the implementation:

Entity	Key Information	Purpose
Profiles / User Records	User ID, email, role	Stores authenticated user profile and role
Repair Requests / Tickets	ID, Client ID, device, issue type, description, status, technician, dates	Stores repair workflow
Notifications	ID, recipient, message, read state, timestamp	Communicates workflow events
Storage / Documents (where used)	File metadata / path	Stores supporting documents/files where required

9. Technical Testing Plan

ID	Test	Input/Action	Expected Result
T01	Client login	Valid credentials	Client dashboard opens
T02	Role routing	Admin/Technician/Client account	Correct dashboard opens
T03	Create repair request	Valid device + issue + description	Request stored with Open status
T04	Admin assignment	Open request + technician	Technician assigned; status Assigned
T05	Technician update	Assigned request	Status/progress updates
T06	Notification	New request/assignment	Correct recipient receives notification
T07	Client tracking	Existing request	Client sees current status
T08	Flutter login	Valid credentials	Flutter client enters application
T09	Database integration	Create/read/update request	Data persists correctly
T10	Invalid access	User attempts restricted function	Access is denied/controlled